HW 7 — Jackson 8.2(b), Eq. 8.94, and 8.14

black = write it down · **red = the answer** · *teal = my note, do not copy* · grey = optional

Q1	Jackson **8.2(b)** — attenuation of a coaxial line	Lec 13
Q2	Derive Jackson **Eq. 8.94** — power lost in a cavity, TM, $p \neq 0$	Lec 15 s12–16
Q3	Jackson **8.14** — graded-index fibre, $n(x) = n(0)\,\mathrm{sech}(\alpha x)$	Lec 16–17

All three follow the lectures directly. Q2 is Lec 15 slides 12–16 (he gets the same result in a shorter notation — the last step converts it to Jackson's form). Q3 is Lec 17 ray tracing — his slide 6 already has the half-period integral this problem asks about.

Everything was checked numerically. Q1: γ from the loss integral matches the closed form to 16 digits. Q2: the conversion $\varepsilon^2\omega^2/\gamma^2 \to (\varepsilon/\mu)[1+(p\pi/\gamma d)^2]$ checked for $p = 0,1,2,3$. Q3: the ray solution satisfies the ray equation to 10^{-16}; Z and L_{opt} were computed by direct integration for three launch angles and agree with π/α and $n(0)Z$ exactly.

Question 1 — Jackson 8.2(b)

> For the coaxial line of 8.2, show that the transmitted power is attenuated as $P(z) = P_0 e^{-2\gamma z}$, with $\gamma = (1/2\sigma\delta)\sqrt{\varepsilon/\mu}\ (1/a + 1/b)/\ln(b/a)$.

ANSWER

$$\gamma = \frac{1}{2\sigma\delta}\sqrt{\frac{\varepsilon}{\mu}}\,\frac{1/a + 1/b}{\ln(b/a)}$$

Flow: *the walls are not perfect, so they eat power → work out how much per metre → divide by the power carried, which part (a) already gave.*

Step 1 — Why the power drops at all

The dielectric is lossless, so the **only** loss is ohmic heating in the two metal walls. A wall carrying a tangential field $H_{\parallel}$ dissipates, per unit area,

$$\frac{dP_{loss}}{dA} = \frac{1}{2\sigma\delta}\,|H_{\parallel}|^2$$ Jackson 8.12

From part (a) the field in the line is $H_\phi = H_0 a/\rho$, so each wall sees its own value.

Step 2 — Add up the two walls

wall	$\lvert H_{\parallel}\rvert$	circumference	loss per metre
inner, $\rho = a$	H_0	$2\pi a$	$\pi a H_0^2/\sigma\delta$
outer, $\rho = b$	$H_0 a/b$	$2\pi b$	$\pi a^2 H_0^2/\sigma\delta b$

Factor a^2 out of both rows:

$$-\frac{dP}{dz} = \frac{\pi a^2 H_0^2}{\sigma\delta}\left(\frac{1}{a} + \frac{1}{b}\right)$$

Step 3 — Turn a loss rate into an exponent

If $P(z) = P_o e^{-2\gamma z}$, differentiating gives $-dP/dz = 2\gamma P$. So

$$\gamma = \frac{-dP/dz}{2P}$$

Divide Step 2 by twice the part (a) power, $P = \sqrt{\mu/\varepsilon}\,\pi a^2 H_o^2 \ln(b/a)$. The whole factor $\pi a^2 H_o^2$ cancels:

$$\gamma = \frac{\pi a^2 H_o^2 (1/a + 1/b)/\sigma\delta}{2\sqrt{\mu/\varepsilon}\,\pi a^2 H_o^2 \ln(b/a)}$$

ANSWER — Q1

$$\gamma = \frac{1}{2\sigma\delta}\sqrt{\varepsilon/\mu}\,\frac{1/a + 1/b}{\ln(b/a)} \quad \checkmark$$

$1/\sqrt{\mu/\varepsilon}$ was written as $\sqrt{\varepsilon/\mu}$.

■ END OF QUESTION 1 ■

Question 2 — Derive Jackson Eq. 8.94

> For a cylindrical cavity of length d, in a **TM mode with p ≠ 0**, show that the power lost in the walls is $P_{loss} = (\varepsilon/\sigma\delta\mu)[1 + (p\pi/\gamma_\lambda d)^2]\,(1 + \xi_\lambda Cd/4A)\int_A|\psi|^2 da$, where C is the circumference and A the cross-sectional area.

ANSWER

$$P_{loss} = \frac{\varepsilon}{\sigma\delta\mu}\left[1 + \left(\frac{p\pi}{\gamma_\lambda d}\right)^2\right]\left(1 + \xi_\lambda \frac{Cd}{4A}\right)\int_A|\psi|^2 da$$

Flow: the cavity has **two kinds of wall** — the sides and the two flat ends. Work out the loss on each separately, add them, then use one standard identity to turn everything into the single integral $\int|\psi|^2 da$.

Step 0 — The field in the cavity

A cavity is a waveguide with two end caps a distance d apart. The ends force a standing wave in z, so the travelling factor e^{ikz} is replaced by a cosine with **$k = p\pi/d$**, p = 0, 1, 2, …

For a **TM** mode the magnetic field is purely transverse:

$$\vec{H} = \vec{H}_t = \frac{i\varepsilon\omega}{\gamma^2}\cos\left(\frac{p\pi z}{d}\right)\hat{z}\times\nabla_t\psi$$ Lec 15, s13

The wall loss formula is the same one as Question 1, now applied over the whole cavity surface:

$$P_{loss} = \frac{1}{2\sigma\delta}\oint\left|\hat{n}\times\vec{H}\right|^2 da$$ Jackson 8.93

Step 1 — Loss on the sides

On a side wall the normal $\hat{n}$ lies in the cross-section. Taking the cross product picks out the **normal derivative** of ψ:

$$\left|\hat{n}\times\vec{H}\right|^2 = \frac{\varepsilon^2\omega^2}{\gamma^4}\left|\frac{\partial\psi}{\partial n}\right|^2\cos^2\left(\frac{p\pi z}{d}\right)$$

The side surface is "perimeter × length", so $da = dL\,dz$. Integrate along z first — a squared cosine over the full length averages to ½:

$$\int_0^d \cos^2\left(\frac{p\pi z}{d}\right)dz = \frac{d}{2}$$ *true for p ≠ 0 — this is where p ≠ 0 is used*

$$P_{sides} = \frac{d}{4\sigma\delta}\frac{\varepsilon^2\omega^2}{\gamma^4}\oint\left|\frac{\partial\psi}{\partial n}\right|^2 dL$$

Step 2 — Loss on the two ends

On an end cap the normal is $\hat{n} = \pm\hat{z}$. Then

$$|\hat{n}\times\vec{H}|^2 = \frac{\varepsilon^2\omega^2}{\gamma^4}\,|\hat{z}\times(\hat{z}\times\nabla_t\psi)|^2 = \frac{\varepsilon^2\omega^2}{\gamma^4}\,|\nabla_t\psi|^2 \qquad \hat{z}\times(\hat{z}\times\vec{A}) = -\vec{A}$$

At the ends $\cos(p\pi z/d)$ is $\cos 0 = 1$ or $\cos p\pi = \pm 1$, so its square is 1 either way. There are **two** ends, which cancels the ½ in front:

$$P_{ends} = \frac{1}{\sigma\delta}\,\frac{\varepsilon^2\omega^2}{\gamma^4}\int_A|\nabla_t\psi|^2 da$$

Use **Green's first identity** exactly as in HW 5 — the boundary term dies because $\psi = 0$ on the wall, and $\nabla_t^2\psi = -\gamma^2\psi$:

$$\int|\nabla_t\psi|^2 da = \gamma^2\int|\psi|^2 da \quad\Rightarrow\quad P_{ends} = \frac{1}{\sigma\delta}\,\frac{\varepsilon^2\omega^2}{\gamma^2}\int_A|\psi|^2 da$$

Step 3 — Add the two

$$P_{loss} = \frac{\varepsilon^2\omega^2}{\sigma\delta\gamma^4}\left[\frac{d}{4}\oint\left|\frac{\partial\psi}{\partial n}\right|^2 dL + \gamma^2\int_A|\psi|^2 da\right]$$

Step 4 — Kill the remaining boundary integral

The one awkward piece is $\oint|\partial\psi/\partial n|^2 dL$. There is a standard identity for it, which simply *defines* the dimensionless number ξ_λ:

$$\oint\left|\frac{\partial\psi}{\partial n}\right|^2 dL = \xi_\lambda\gamma^2\,\frac{C}{A}\int_A|\psi|^2 da$$ Jackson 8.62 · Lec 15 s16

Substitute it in. Every term now carries the same $\int|\psi|^2 da$, so it factors out:

$$P_{loss} = \frac{\varepsilon^2\omega^2}{\sigma\delta\gamma^4}\left[\frac{d}{4}\xi_\lambda\gamma^2\,\frac{C}{A} + \gamma^2\right]\int|\psi|^2 da$$

One γ^2 cancels:

$$P_{loss} = \frac{\varepsilon^2\omega^2}{\sigma\delta\gamma^2}\left(1 + \xi_\lambda\,\frac{Cd}{4A}\right)\int_A|\psi|^2 da$$ = Lec 15, s16

Step 5 — Put it in Jackson's form

The answer above is complete, but Jackson writes the prefactor differently. The two are the same thing — here is the conversion.

In a cavity the resonance condition is

$$\mu\varepsilon\omega^2 = \gamma_\lambda^2 + \left(\frac{p\pi}{d}\right)^2$$ *transverse part + z part* Lec 14, s14

So solve that for $\varepsilon\omega^2$ and substitute:

$$\frac{\varepsilon^2\omega^2}{\gamma_\lambda^2} = \varepsilon\,\frac{\varepsilon\omega^2}{\gamma_\lambda^2} = \varepsilon\,\frac{\gamma_\lambda^2 + (p\pi/d)^2}{\mu\,\gamma_\lambda^2}$$

Split the fraction:

$$= \frac{\varepsilon}{\mu}\left[1 + \left(\frac{p\pi}{\gamma_\lambda d}\right)^2\right]$$

Checked numerically for p = 0, 1, 2, 3 — the two expressions agree to 16 digits every time.

ANSWER — Q2 (Eq. 8.94)

$$P_{loss} = \frac{\varepsilon}{\sigma\delta\mu}\left[1 + \left(\frac{p\pi}{\gamma_\lambda d}\right)^2\right]\left(1 + \xi_\lambda\,\frac{Cd}{4A}\right)\int_A |\psi|^2 da \quad \checkmark$$

The $[1 + (p\pi/\gamma_\lambda d)^2]$ is the standing wave in z; the $(1 + \xi_\lambda Cd/4A)$ is the sides compared with the ends.

■ END OF QUESTION 2 ■

OPTIONAL For $p = 0$ the z-integral in Step 1 gives d, not d/2, so the sides term doubles — which is why Jackson says "for $p = 0$, ξ_λ must be replaced by $2\xi_\lambda$". The bracket $[1 + (p\pi/\gamma d)^2]$ becomes 1.

Question 3 — Jackson 8.14

> An optical fibre has a graded index $n(x) = n(0)\,\mathrm{sech}(\alpha x)$, for rays in the x–z plane. The invariant is $\bar{n} = n(x_{max}) = n(0)\cos\theta(0)$. **(a)** Solve the eikonal equations for x(z) and show $\alpha x = \sinh^{-1}[\sinh(\alpha x_{max})\sin(\alpha z)]$; sketch rays for $\theta(0) = \pi/6, \pi/4, \pi/3$. **(b)** Find the half-period Z. Does it depend on $\bar{n}$? **(c)** Show $L_{opt} = \int n(x)ds = n(0)Z$, and comment.

ANSWER

(a) $\sinh(\alpha x) = \sinh(\alpha x_{max})\sin(\alpha z)$

(b) $Z = \dfrac{\pi}{\alpha}$ — **independent** of $\bar{n}$

(c) $L_{opt} = n(0)\,Z$ — same for every ray

Flow: the ray equation gives dx/dz → separate and integrate → the substitution $u = \sinh(\alpha x)$ turns it into a plain arcsine → then (b) and (c) are read off that one result.

Step 1 — The ray equation for a slab

For an index that varies only with x, ray tracing gives a first integral — the quantity $\bar{n}$ is conserved along the ray:

$$\bar{n}^2\left(\frac{dx}{dz}\right)^2 = n^2 - \bar{n}^2$$ Lec 17, s5

Take the square root and separate:

$$\frac{dx}{dz} = \frac{\sqrt{n^2 - \bar{n}^2}}{n} \quad\Rightarrow\quad dz = \bar{n}\,\frac{dx}{\sqrt{n^2 - \bar{n}^2}}$$

At the turning point $x = x_{max}$ the ray is momentarily flat, $dx/dz = 0$, so $n(x_{max}) = \bar{n}$ — that is exactly the invariant the problem defines.

Step 2 — Substitute $u = \sinh(\alpha x)$

Put in $n = n_o \operatorname{sech}(\alpha x)$, writing $n_o = n(0)$. The substitution that clears the sech is **$u = \sinh(\alpha x)$**, because $\cosh^2 = 1 + \sinh^2$:

$$n^2 = \frac{n_o^2}{\cosh^2(\alpha x)} = \frac{n_o^2}{1 + u^2}, \qquad du = \alpha \cosh(\alpha x)\, dx \;\Rightarrow\; dx = \frac{du}{\alpha\sqrt{1+u^2}}$$

The messy square root becomes

$$\overline{n^2} - \overline{n}^2 = \frac{n_o^2 - n^2(1+u^2)}{1 + u^2}$$

Now put both into dz. The $\sqrt{1+u^2}$ factors **cancel**:

$$dz = \frac{n\, du}{\alpha\sqrt{n_o^2 - n^2 - n^2 u^2}}$$

Step 3 — Integrate — it is an arcsine

Pull n out of the root so it reads $\int du/\sqrt{U^2-u^2}$, the standard arcsine form, where

$$U \equiv \frac{\sqrt{n_o^2 - n^2}}{n}$$

$$z = \frac{1}{\alpha}\int \frac{du}{\sqrt{U^2 - u^2}} = \frac{1}{\alpha}\sin^{-1}\frac{u}{U}$$ *the problem puts $z = 0$ at $x = 0$, so the constant is zero*

Invert it:

$$u = U\sin(\alpha z) \quad \Rightarrow \quad \sinh(\alpha x) = U\sin(\alpha z)$$

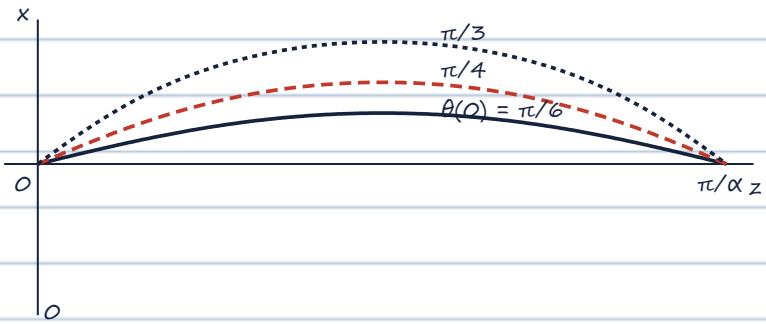
Step 4 — Recognise U

From Step 1, at the turning point $n(x_{max}) = n$, so $n_o\,\mathrm{sech}(\alpha x_{max}) = n$, i.e. $\cosh(\alpha x_{max}) = n_o/n$. Then

$$\sinh(\alpha x_{max}) = \sqrt{\cosh^2 - 1} = \sqrt{n_o^2/n^2 - 1} = \frac{\sqrt{n_o^2 - n^2}}{n} = U$$

ANSWER — (a)

$$\sinh(\alpha x) = \sinh(\alpha x_{max})\sin(\alpha z) \quad \text{i.e.} \quad \alpha x = \sinh^{-1}[\sinh(\alpha x_{max})\sin(\alpha z)] \;\checkmark$$



*One half-period. A bigger launch angle swings further out, but **every** ray crosses the axis again at the same $z = \pi/\alpha$.*

PART (b) — the half-period Z

A half-period ends when the ray comes back to the axis, $x = 0$. From part (a) that needs $\sin(\alpha z) = 0$, and the first non-zero solution is $\alpha z = \pi$:

ANSWER — (b)

$$Z = \frac{\pi}{\alpha}$$ — it does **not** depend on $\bar{n}$

α is the only quantity in it. Every ray, whatever its launch angle, returns to the axis after the same distance.

PART (c) — optical path length

Step 1 — Turn ds into dz

Along the ray, $ds = \sqrt{dx^2+dz^2} = dz\sqrt{1 + (dx/dz)^2}$. Put in dx/dz from Step 1 of part (a):

$$1 + \left(\frac{dx}{dz}\right)^2 = 1 + \frac{n^2-n^2}{n^2} = \frac{n^2}{n^2} \quad \Rightarrow \quad ds = \frac{n}{n}\,dz$$

$$L_{opt} = \int n\,ds = \frac{1}{n}\int_0^z n^2\,dz \qquad \textit{everything is now in z alone}$$

Step 2 — Write n^2 using part (a)

With $S \equiv \sinh(\alpha x_{max})$, part (a) says $\sinh(\alpha x) = S\sin(\alpha z)$, so $\cosh^2(\alpha x) = 1 + S^2\sin^2(\alpha z)$ and

$$n^2 = \frac{n_o^2}{1 + S^2\sin^2(\alpha z)}$$

$$L_{opt} = \frac{n_o^2}{n}\int_0^{\pi/\alpha} \frac{dz}{1 + S^2\sin^2(\alpha z)}$$

Step 3 — Use the hint

Substitute $t = \alpha z$ (so $dz = dt/\alpha$, t runs $0 \to \pi$), then fold the symmetric halves together to get $2\int_0^{\pi/2}$. The given integral is

$$\int_0^{\pi/2} \frac{dt}{1 + S^2\sin^2 t} = \frac{\pi}{2\sqrt{1+S^2}} \qquad \text{the hint in the question}$$

$$\Rightarrow L_{opt} = \frac{n_0^2}{n\alpha} \cdot 2 \cdot \frac{\pi}{2\sqrt{1+S^2}} = \frac{\pi n_0^2}{n\alpha\sqrt{1+S^2}}$$

Finally, $\sqrt{1+S^2} = \cosh(\alpha x_{max}) = n_0/\bar{n}$ from Step 4 of part (a). Substituting, the $\bar{n}$ cancels and one n_0 cancels:

ANSWER — (c)

$$L_{opt} = \frac{\pi n_0}{\alpha} = n(0)\,Z \quad \checkmark$$

Comment. *L_{opt} contains **no** $\bar{n}$, so every ray has the same optical path length per half-period, no matter what angle it was launched at. Optical path length is what sets travel time, so **all rays arrive together** — the pulse does not smear out. That is exactly what a multimode fibre wants, and it is why the sech profile is the ideal grading. A step-index fibre has no such property: steep rays take visibly longer, which limits how fast you can send data.*

■ END OF QUESTION 3 ■